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Task 2: House Price Prediction Using Linear Regression

1. Introduction

Linear Regression is a supervised machine learning algorithm used to predict continuous values based on input features. It establishes a linear relationship between independent and dependent variables. In this task, Linear Regression is used to predict house prices based on the size of the house using Python.

2. Objective

The objectives of this task are:

- To understand the concept of Linear Regression
 - To implement Linear Regression using Python
 - To predict house prices based on house size
 - To visualize the regression model using a graph
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3. Tools and Technologies Used

- **Programming Language:** Python
 - **Libraries Used:**
 - Pandas – for data handling
 - Matplotlib – for data visualization
 - Scikit-learn – for implementing Linear Regression
 - **IDE:** Visual Studio Code
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4. Dataset Description

The dataset used in this task is house_prices.csv.
It contains two columns:

- **Size** – size of the house in square feet
- **Price** – price of the house in Indian Rupees

Each row represents a house with its corresponding size and price.

5. Methodology

1. The dataset was loaded using the Pandas `read_csv()` function.
 2. The independent variable (Size) and dependent variable (Price) were selected.
 3. A Linear Regression model was created using the `LinearRegression()` class from `scikit-learn`.
 4. The model was trained using the available dataset.
 5. House price was predicted for a given house size (1600 sq ft).
 6. A scatter plot and regression line were generated to visualize the relationship between size and price.
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6. Results and Observations

- The Linear Regression model successfully predicted the house price for a house size of **1600 sq ft**.
 - The predicted price obtained was approximately **₹23,65,543**.
 - The regression line shows a positive linear relationship between house size and price.
 - As the house size increases, the predicted house price also increases.
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7. Conclusion

This task demonstrated the practical implementation of Linear Regression using Python. The model effectively predicted house prices based on house size. This task helped in understanding basic machine learning concepts, data preprocessing, model training, and result visualization. Linear Regression is a simple yet powerful algorithm for predictive analysis.

8. Output

- Python program (`house_price_prediction.py`)
- Dataset file (`house_prices.csv`)
- Regression graph showing actual data points and predicted line